Hands-On Tasks for Linux Boot Process

Task 1: Check Current Kernel Version

Objective: Identify the running Linux kernel.

Command

uname -r

Expected Output Example

5.14.0-362.el9.x86_64

Learning:
This shows the **kernel currently loaded during the boot process**.

Task 2: View Kernel Files in /boot

Objective: Locate kernel and initramfs files used during boot.

Command

ls /boot

Look for Files Like

vmlinuz-5.14.0
initramfs-5.14.0.img
grub

Learning

- `vmlinuz` → Linux kernel
- `initramfs` → temporary filesystem used during boot

Task 3: Check Bootloader Configuration

Objective: View GRUB bootloader configuration.

Command

cat /boot/grub2/grub.cfg

Or on Ubuntu

cat /boot/grub/grub.cfg

Learning
This file contains:

- boot entries
- kernel paths
- boot parameters

Task 4: Check Default GRUB Settings

Objective: View default boot configuration.

Command

cat /etc/default/grub

Example

GRUB_TIMEOUT=5
GRUB_DEFAULT=saved
GRUB_CMDLINE_LINUX="quiet"

Learning
Controls:

- boot timeout
- default OS selection

- kernel parameters

Task 5: Check First Process (PID 1)

Objective: Identify the first process started by the kernel.

Command

ps -p 1

Expected Output

PID TTY TIME CMD
1 ? 00:00 systemd

Learning

- `systemd` is the **first user-space process**
- Always **PID = 1**

Task 6: Check System Boot Time

Objective: Check how long the system took to boot.

Command

systemd-analyze

Example Output

Startup finished in 2.1s (kernel) + 3.5s (userspace) = 5.6s

Learning
Shows **kernel boot time and userspace boot time**.

Task 7: View Boot Performance Details

Objective: Identify slow services during boot.

Command

systemd-analyze blame

Example

2.321s NetworkManager.service
1.932s firewalld.service

Learning
Shows **which services take longest during startup**.

Task 8: Check Default Boot Target

Objective: Identify the system startup mode.

Command

systemctl get-default

Output Example

graphical.target

Learning

Target	Meaning
multi-user.target	CLI mode
graphical.target	GUI mode

Task 9: Change Boot Target

Objective: Change system startup mode.

Set CLI Mode

sudo systemctl set-default multi-user.target

Set GUI Mode

sudo systemctl set-default graphical.target

Verify

systemctl get-default

Task 10: Check Running Services

Objective: View services started during boot.

Command

systemctl list-units --type=service

Learning
Shows services like:

- sshd
- crond
- NetworkManager
- firewalld

Task 11: View Boot Logs

Objective: Check system boot logs.

Command

journalctl -b

Learning
Displays **logs generated during the current boot process**.

Task 12: Check Previous Boot Logs

Objective: View logs from previous boot.

Command

journalctl -b -1

Learning
Helps in **boot troubleshooting**.

Task 13: Identify Boot Device

Objective: Check disk partitions.

Command

lsblk

Example

sda
├─sda1 /boot
├─sda2 /

Learning
Shows where **boot and root filesystem are located**.

Task 14: Check Mounted Filesystems

Objective: Verify root filesystem.

Command

df -h

Look for:

/dev/sda2 /
